

When usage exceeds capacity

Each directed M1 or M2 track has capacity one on tiny_dr

The idea

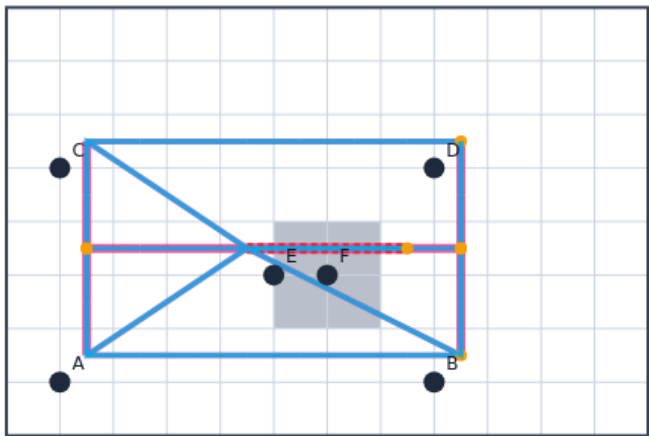
- Path_track_usage walks consecutive segment pairs
- Track_overflow computes max of zero and usage minus capacity per track
- Sequential L-HV on all six nets should yield positive total overflow at cap one

Usage heat

TRACK USAGE

Usage heat

Step 1 / 5 · def · module02-05-track-usage



Explanation

Each M1/M2 track accumulates routed net count.

- Heat on canvas
- Red = overflow

Usage map

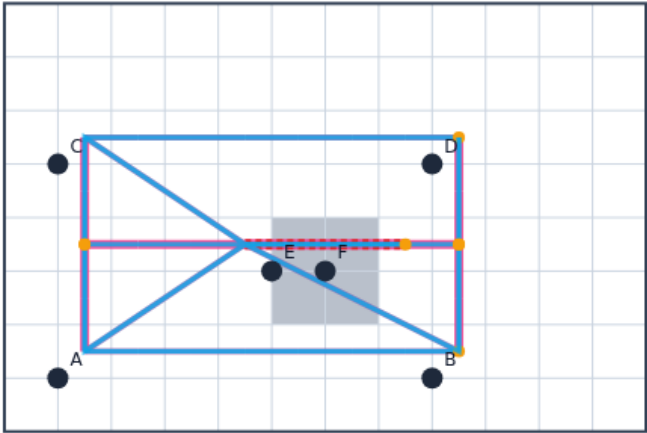
Usage heat

Total overflow

TRACK USAGE

Total overflow

Step 2 / 5 · total · module02-05-track-usage



Explanation

Sum over tracks—primary scalar.

- Primary metric

spread layered≈3

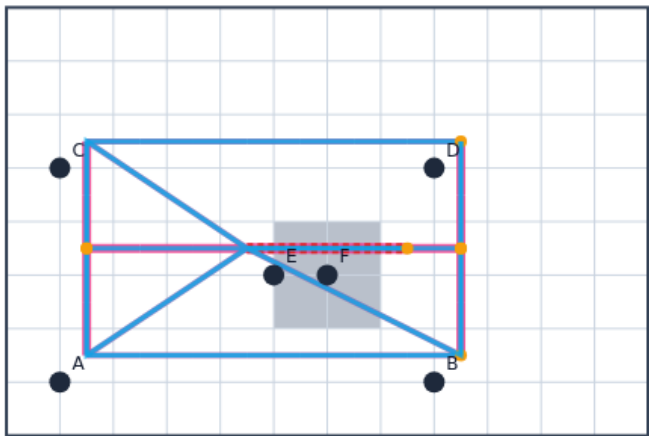
Total overflow

Max overflow

TRACK USAGE

Max overflow

Step 3 / 5 · max · module02-05-track-usage



Explanation

Worst track catches hotspot corridors.

- Hotspot

max≈1

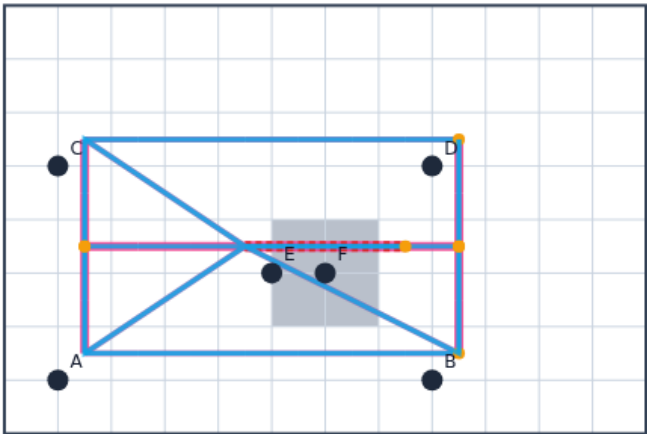
Max overflow

Overflow count

TRACK USAGE

Overflow count

Step 4 / 5 · count · module02-05-track-usage



Explanation

How many tracks exceed Cap.

- Triple report

Same pattern as global

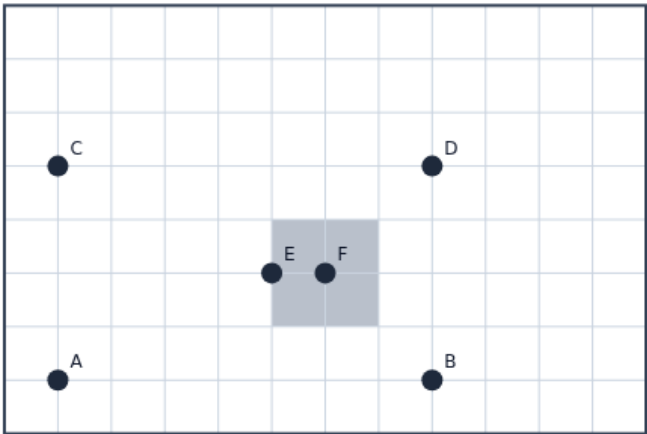
Overflow count

Hit targets

TRACK USAGE

Hit targets

Step 5 / 5 · target · module02-05-track-usage



Explanation

Move pins or switch Lee/A* to clear thresholds.

- Challenge lab

No golden mode

Hit targets

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Track usage heat

INTERACTIVE TOOL

Track usage heat

Report total, max, and overflowing-track count from routed usage.

Your job: route layered on spread (total ≈ 3), then reduce overflow by moving cells or switching to A*.

Reset workspace

Challenges0 / 8 cleared

Pick one[Intro] Spread layered overflow

Pick a challenge and click **Start**. Move cells, route, then **Check**.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A B C D E F←→↑↓Route LeeRoute layeredRoute A*

Rip-up detailed

Graph

Reference / metrics

selected: A @ grid (1,1)
routes: (none – click Route)
overflow total: 0.00 · max: 0.00 · count: 0
track capacity: 2 · mode: astar
HPWL: 34.0

Browser lab starter

learn_routing — 05 track usage

Implement track

- Implement ``path_track_usage`` and ``track_overflow``
- Call `sequential_detailed` with mode `I_hv` on `tiny_dr` and assert total overflow is greater

Pitfalls

- Computing overflow before summing all nets
- Using GCell edge usage from global routing instead of directed tracks
- Reporting negative overflow values

Your turn

- Hit overflow targets
- Next: assign vias on two-layer L paths